KAIZEN MATHS

Lesson starter

Permutations and Combinations

Combinations & Probability

Name _____

SA1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

How many ways can 6 questions be selected from a bank of 18 questions?

SA2

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A committee must have 3 men and 3 women. There are 8 men and 7 women available. How many committees are possible?

Show your working

KAIZEN MATHS

Lesson starter

Permutations and Combinations

Combinations & Probability

Name _____

SB1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A bag contains 7 red counters and 7 blue counters. 4 counters are chosen without replacement. Find the probability of choosing exactly 2 red and 2 blue counters.

SB2

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

Find the coefficient of x^3 in the expansion of $(1 + x)^8$.

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Combinations & Probability

Name

EA1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

How many ways can 4 questions be selected from a bank of 10 questions?

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Combinations & Probability

Name

EB1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A committee must have 4 men and 4 women. There are 8 men and 6 women available. How many committees are possible?

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Combinations & Probability

Name

EC1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A bag contains 5 red counters and 5 blue counters. 4 counters are chosen without replacement. Find the probability of choosing exactly 1 red and 3 blue counters.

Show your working

KAIZEN MATHS

Exit ticket

Permutations and Combinations

Combinations & Probability

Name

EA1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

How many ways can 4 questions be selected from a bank of 10 questions?

Show your working

Teacher answers & working

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SA1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

How many ways can 6 questions be selected from a bank of 18 questions?

Answer:

18,564

Working:

This is a selection, not an arrangement, so use combinations.

${}^{18}C_6 = \frac{18!}{6!12!}$

${}^{18}C_6 = 18{,}564$

SA2

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A committee must have 3 men and 3 women. There are 8 men and 7 women available. How many committees are possible?

Answer:

1,960

Working:

Choose the men and women separately, then multiply the choices.

Ways to choose the men = ${}^{8}C_3 = 56$

Ways to choose the women = ${}^{7}C_3 = 35$

Total committees = 56 × 35 = 1,960

SB1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A bag contains 7 red counters and 7 blue counters. 4 counters are chosen without replacement. Find the probability of choosing exactly 2 red and 2 blue counters.

Answer:

$\frac{63}{143}$

Working:

Count the favourable selections and the total selections.

Favourable selections = ${}^{7}C_2 \times {}^{7}C_2$

Favourable selections = 21 × 21 = 441

Total selections = ${}^{14}C_4 = 1{,}001$

Probability = $\frac{441}{1{,}001} = \frac{63}{143}$

Teacher answers & working

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SB2

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

Find the coefficient of x^3 in the expansion of $(1 + x)^8$.

Answer:

56

Working:

In $(1 + x)^n$, the coefficient of x^r is nCr.

Coefficient = ${}^8C_3$

${}^8C_3 = \frac{8!}{3!5!} = 56$

EA1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

How many ways can 4 questions be selected from a bank of 10 questions?

Answer:

210

Working:

This is a selection, not an arrangement, so use combinations.

${}^{10}C_4 = \frac{10!}{4!6!}$

${}^{10}C_4 = 210$

EB1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A committee must have 4 men and 4 women. There are 8 men and 6 women available. How many committees are possible?

Answer:

1,050

Working:

Choose the men and women separately, then multiply the choices.

Ways to choose the men = ${}^8C_4 = 70$

Ways to choose the women = ${}^6C_4 = 15$

Total committees = 70 × 15 = 1,050

Teacher answers & working

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EC1

Use combinations when order does not matter. For probability, divide favourable outcomes by total outcomes.

A bag contains 5 red counters and 5 blue counters. 4 counters are chosen without replacement. Find the probability of choosing exactly 1 red and 3 blue counters.

Answer:

$\frac{5}{21}$

Working:

Count the favourable selections and the total selections.

Favourable selections = ${}^{5}C_{1} \times {}^{5}C_{3}$

Favourable selections = $5 \times 10 = 50$

Total selections = ${}^{10}C_{4} = 210$

Probability = $\frac{50}{210} = \frac{5}{21}$